

ML-guided ASIC TM

ABC Technology Mapping

ASIC TM steps of ABC

- Compute k -feasible cuts (ABC uses $k = 5$)
- Compute truth-tables of cuts
- Boolean Matching
- Compute best arrival time at each node
- Choose the best cover

Problem: The number of cuts available per node is exponential in the number of cut leaves and size of the input AIG

Solution:

- Prune the number of cuts heuristically
- Heuristics are mostly handcrafted through experimental analysis and based on domain-specific knowledge – based on static attributes of the cut/node e.g., number of leaves, cut level, cut volume

Note: Run-time efficiency of TM-algorithm can be estimated based on the number of cuts exposed to the mapper

Goal: To develop an effective cut-based heuristic to achieve very similar results to the exhaustive approach while preserving a small number of utilized cuts

DSE of Cut-based ASIC TM

Objective: Evaluate and show the impacts of cut-sorting and filtering heuristics

Experimental setup

- Only modify the cut sorting and filtering heuristic of ABC TM
- ABC keeps the top 250 cuts per node after sorting the cuts by their number of leaves and ruling out dominated cuts

Observation: After sorting the cuts on each node by a different attributes (other than the number of leaves) did not show consistent benefit

Lesson learned:

- Depending upon each node's context, the sorting policy might be different
- Different structural features from the cut should be considered simultaneously while evaluating good cuts

DSE of Cut-based ASIC TM

What about randomly shuffling the cut list?

Q: Why ordering of cuts matter?

A: The choice of one cut implies other cuts must be chosen

Experiment: After disabling the filtering of dominated cuts, the cut list of each node is randomly shuffled

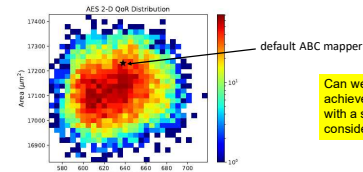
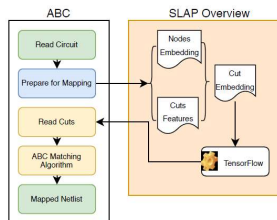


Fig. 1. 2-D QoR Distribution for an AES. Each data-point indicates the QoR of one mapping-solution by randomly shuffling the list of cuts for each node. The black star indicates the default ABC mapping result.

SLAP: Supervised Learning Approach for Priority-cuts TM

Overview

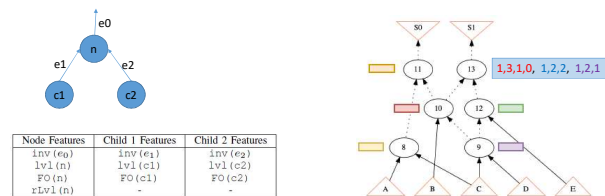
- SLAP uses an ML model to co-relate multiple structural features of cuts and serve as a replacement for pruning heuristics.
- The used features are chosen to capture a sense of the graph structure on which the cut appears.
- Timing information was used during training to learn cut filtering policies for delay optimization.



SLAP

Node embedding

- Captures the node structural information by considering features from the node and its children



SLAP

Cut embedding

- Collects structural features for each cut

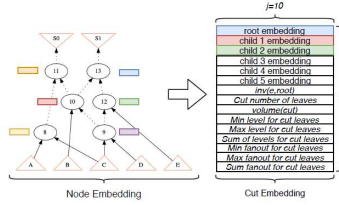


Fig. 2. The left-hand DAG represents an AIG, where each colored box represents the node embeddings. These node embeddings along with the cut features collected are used to derive the cut embedding. The first 5-rows represent the node embeddings from the root node and the cut leaves, followed by the cut features.

Overall Model and Training

Training data

- For a given circuit, different tech-mapped results are generated by changing the cut-sorting and filtering heuristics
- Label each cut used in a tech-mapping with normalized delay QoR
- Formally,
 - Let S be the set of all the unique solutions possible for mapping one circuit.
 - Let $m \in S$ be one valid mapping for the circuit, and C be the set of cuts used to implement m .
 - Then, each $c \in C$ has as label the timing information of the implementation m normalized to the largest delay available at S .

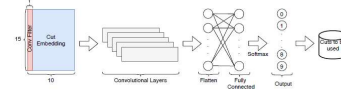


Fig. 3. Overview of the proposed model.

Classifier: multi-class classifier that labels a cut in one out of 10 distinct QoR classes (Class-0: min delay, Class-9: max delay)

A CNN layer having 128 filters (15 x 1) with stride = 1 is used to correlate the node features with the cut features

Conv. layer generates 128 tensors having dimension 10x1 each, which is passed through a flatten layer, generating a layer of 1280 neurons, which is fully connected to a dense layer of 10 neurons (use softmax to predict the output class)

Loss function: sparse categorical cross-entropy

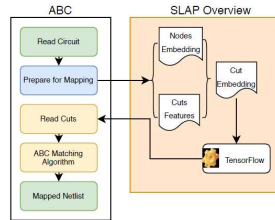
SLAP: Putting it all together

ML-guided cut filtering

- For each node n_i , the list of k feasible cuts is L_i
- For each cut $c \in L_i$, the model outputs the class
- Based on experiments cuts are divided into three classes
 - Good cuts (0-3)
 - Average cuts (4-6)
 - Bad cuts (7-9)
- Cut filtering
 - Expose only good cuts to the mapper, if available
 - If there is no good cut, expose average cuts only
 - Otherwise, expose the trivial cut only

A note on model accuracy:

- predicting the QoR class of a cut for 10 classes has an accuracy of around 34.01% on the validation set
- However, we are interested in considering the cuts classified to be in the top x QoR classes ($x = 6$)
- When we convert the problem to a binary-classifier (classifies whether the cut should be considered during tech-mapping or not) the model accuracy is 93.4% on the validation set



SLAP: Results

Circuit	ABC Original			ABC Unlimited			SLAP		
	Area (μm^2)	Delay (ps)	Cuts Used	Area (μm^2)	Delay (ps)	Cuts Used	Area (μm^2)	Delay (ps)	Cuts Used
adder	898.13	3,770.65	10,118	1,009.40	3,404.62	18,228	1,031.33	3,268.67	1,352
bar	2,680.39	1,114.90	42,760	2,680.39	1,114.90	43,784	3,083.23	923.82	10,705
ce288	3,000.45	1,265.88	95,519	3,014.68	1,228.32	197,022	3,023.54	1,236.49	111,409
max	2,312.27	3,809.03	39,727	2,049.60	3,980.21	41,478	2,292.44	3,710.54	22,521
rc256b	1,794.39	7,388.03	22,387	2,482.33	6,861.76	40,978	2,428.21	6,807.02	30,476
rc64b	450.70	1,844.59	5,491	601.16	1,688.38	10,066	602.33	1,565.70	6,397
sin	5,207.04	3,955.57	191,131	5,073.14	3,737.30	290,664	5,087.6	3,584.79	141,788
c7552	2,045.40	817.46	52,150	1,905.90	828.46	93,745	2,002.01	800.09	30,933
mul32-boolh	6,802.44	1,837.68	185,406	5,623.21	1,743.56	355,106	5,597.55	1,772.47	147,043
mul64-boolh	25,717.95	3,583.35	733,156	20,797.85	3,743.95	1,394,922	21,984.07	3,580.47	601,889
square	15,744.07	3,680.87	541,321	14,107.84	2,970.49	919,522	15,789.09	3,023.01	380,787
AES	17,321.27	638.09	264,380	16,994.21	632.07	317,827	16,489.63	594.64	145,532
64b_mult	25,458.31	4,649.10	833,565	24,168.27	4,144.31	1,484,104	24,021.07	4,278.48	892,650
Pico RISCv	12,190.05	1,782.08	181,410	12,010.19	2,126.47	197,069	12,148.52	1,601.21	103,174
Geomean	4,637.80	2,297.91	96,321.81	4,610.87	2,222.83	150,657.67	4,760.47	2,090.01	73,739.02
Improvements	1.0	1.0	1.0	0.99	0.96	1.56	1.02	0.96	0.76

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Δ SLAP/ABC			Δ SLAP/ABC (Unlimited)		
Area	Delay	Cuts	Area	Delay	Cuts
1.15	0.97	1.34	1.02	0.98	0.74
1.15	0.83	0.25	1.15	0.83	0.24
1.00	0.98	1.17	1.00	1.01	0.57
0.99	0.97	0.57	1.12	0.93	0.54
1.15	0.92	1.06	0.98	0.99	0.74
1.34	0.85	1.16	1.00	0.93	0.64
0.98	0.91	0.74	1.00	0.98	0.49
0.98	0.98	0.59	1.05	0.97	0.35
0.82	0.97	0.79	1.00	1.02	0.41
0.85	0.92	0.82	1.06	0.88	0.45
1.00	0.82	0.70	1.12	1.02	0.41
0.95	0.93	0.55	0.97	0.94	0.46
0.94	0.92	1.07	0.99	1.03	0.60
1.00	0.90	0.57	1.01	0.75	0.52
1.01	0.90	0.77	1.01	0.84	0.42